A4 past final worksheet

1. (Fall 2024, #3) Determine whether the following integrals are convergent or divergent. Justify your answer.

   (a) $\displaystyle\int_0^1 \frac{1+e^{-x}}{x^3}dx$

   (b) $\displaystyle\int_1^{+\infty} \frac{2x}{x^3+1}dx$

2. (Spring 2024, #2) Find the following antiderivatives.

(a) $\displaystyle\int \frac{\sqrt{4-x^2}}{x}dx$

(b) $\displaystyle\frac{x+1}{x^3+x}dx$

3. (Spring 2024, #3) Consider the improper integral

$$\int_0^1 \frac{5 - \sin x}{x^{1/3} + x^3} dx.$$

A student, investigating this integral, makes the following claims:

I. $\dfrac{5 - \sin x}{x^{1/3} + x^3} \le \dfrac{6}{x^3}$ when $0 < x \le 1$

II. $\displaystyle\int_0^1 \frac{6}{x^3} dx$ is convergent.

III. It follows from Claims I and II that the integral is convergent.

(a) Identify the flaw in the student's argument. Circle the best response.

i. Claim I is false.

ii. Claim I is true but claim II is false.

iii. Claims I and II are true but it does not follow from them that the integral is convergent.

(b) Is the integral convergent of divergent? Carefully justify your answer.

4. (Spring 2023, #5) Evaluate the following improper integral:

$$\int_5^{\infty} \frac{3}{x^2 + x} dx$$

Hint: the integral is known to converge.

5. (Fall 2021, #2) Compute the following integrals. If an integral is determined to be improper, you must also justify its convergence.

(a) $\displaystyle\int_0^1 \frac{x}{\sqrt{1-(x-1)^2}}dx$

(b) $\displaystyle\int_0^1 x^{1/4}\ln x dx$